

Kamal Joshi

Staff Software Engineer | Infrastructure-as-Code | Go | Angular | AWS | Distributed Systems

Austin, TX | Open to hybrid

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PROFESSIONAL SUMMARY

Staff software engineer and technical leader with deep experience building robust, scalable infrastructure platforms, distributed systems, and automation services in high-availability, security-critical environments. Deep hands-on expertise in infrastructure-as-code — designing and delivering multi-region, multi-account Terraform pipelines and automation services in production. Proven track record of operating autonomously, translating ambiguous requirements into durable technical solutions, and influencing engineering standards and design decisions across teams. I use AI-powered tools proficiently in my daily workflow — for coding, debugging, testing, and refactoring — while maintaining engineering rigor and verifying outputs for security, performance, and accuracy.

CORE STRENGTHS

- **Infrastructure-as-Code:** Terraform (modules, pipelines, multi-account, multi-region), staged rollouts, sovereign/GovCloud environments, Kubernetes, Docker, CI/CD
- **Distributed systems & automation services:** Go, Java, Spring Boot, gRPC, REST/JSON APIs, event-driven microservices, high-availability architecture
- **AWS:** Lambda, ECS, EKS, S3, SQS/SNS, DynamoDB, Secrets Manager — certified and production-operated
- **Security & compliance:** OAuth 2.0, OIDC, API key management, secure credential management, audit logging; led ISMAP security certification
- **Angular & front-end:** Angular, TypeScript, JavaScript — built production UI features including search and dashboard components
- **Technical leadership:** design reviews, code reviews, mentoring, cross-functional Agile/Scrum collaboration, engineering standards
- **AI-powered development:** daily use of Claude Code, GitHub Copilot, and Codex — coding, debugging, testing, refactoring; pioneered team-wide AI tooling adoption

EXPERIENCE

Cisco

Software Engineering Technical Leader

November 2025 - Present | Austin, TX

- Delivered infrastructure-as-code platform changes across sovereign, GovCloud, and multi-region environments, decomposing a high-risk Terraform rollout into staged changes across ECR, Secrets Manager, DynamoDB streams, SNS, service config, and audit logging — validating high-availability behavior at each step.
- Designed and owned automation services for Cisco Secure Access's **API Gateway, Authorization Service, and Third-Party Integrations Service**, building scalable, reliable infrastructure management products across multi-region deployments.
- Translated ambiguous requirements into a robust technical solution by reframing API onboarding as a control-plane architecture problem — proposing a hybrid Kong Admin API + Terraform model that influenced the product roadmap.
- Maintained high-availability and security requirements under production pressure: identified a **300k+ message DLQ backlog** across **15 organizations**, executed a full redrive within **5-10 minutes** with zero data loss.
- Resolved a customer-scale infrastructure failure by tracing **250+ appliance registration requests** through API key authentication and rate-limit behavior, driving durable fixes across retry/backoff, token reuse, and service ownership.

- Led cross-service design for **Splunk** and **CloudEvents** delivery formats, coordinating four Agile teams to a shared technical contract — contributing to scalable, reusable service architecture.
- Mapped AuthSvc TLS secret synchronization, API key refresh behavior, and access control flows across Kubernetes, Secrets Manager, and Kong — reducing ambiguity for security-sensitive IaC automation.

Senior Software Engineer

January 2021 - November 2025 | Austin, TX

- Designed and deployed **OPA (Open Policy Agent)** as a sidecar for policy enforcement — building security controls directly into platform services.
- Led **ISMAP security certification** for the APIX platform end-to-end, coordinating cross-functional compliance requirements and meeting all deadlines.
- Pioneered team-wide **GitHub Copilot** adoption ahead of broader Cisco AI initiatives; applied AI-powered tools (Claude Code, Codex) for coding, debugging, testing, and refactoring with a verification-first discipline.
- Mentored early-in-career engineers through design reviews, code reviews, and team planning; initiated **Tip Talk Tue**, a recurring knowledge-sharing forum on engineering productivity and security practices.

Randstad Technologies US at Cisco

Senior Software Developer

March 2015 - January 2021 | Dublin, OH

- Designed and implemented **Go-based gRPC and JSON/HTTP APIs** for a cloud-native data ingestion platform; built OIDC/OAuth 2.0 authentication and authorization flows.
- Built **Angular** applications and UI features for threat awareness and security analytics products, including a type-assisted search feature that reduced friction for security analysts.
- Contributed to automatic generation of protocol buffer specs and gRPC APIs from SQL schemas, improving API consistency across services.
- Reduced **100+ OWASP-reported security warnings to zero** through secure coding, browser security headers, input sanitization, and server-side validation.

Earlier Engineering Experience

Cognizant, Oracle, Collabera | December 2007 - February 2015

- Delivered backend services, workflow applications, and integration features across financial services, healthcare, insurance, and public-sector domains.
- Built a foundation in Java, REST APIs, service integrations, and scalable engineering practices.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- MongoDB Certified Java Developer M101J
- Google Cloud Platform Essentials

EDUCATION

Bachelor of Engineering, Information Technology
Dharmsinh Desai University